

经 典 电 动 力 学

清华大学物理系

王 青









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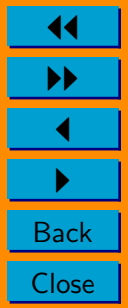
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第4章　电磁波的辐射第456节　25个问题驱动的互动学习 =====⇒问题号题目(得票)

问题1 （4月21日周一讨论）：9-2多极矩的图像（26）； 10-2辐射的多极展开（26）；

问题2 （4月23日周三讨论）：10-8卷积（35）；

10-5磁多极辐射和电多极辐射（25）； 9-8与转动惯量、引力的类比（24）；

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9-9基本粒子的电偶极矩与时间反演（22）； 10-1辐射（22）； 10-4天线（22）；

10-3辐射多极展开的条件（20）； 10-14惠更斯原理（19）；

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10-10光束定义（15）； 9-5磁矢势零阶项（14）； 9-7多极展开的条件（14）

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10-16惠更斯原理的体边对应（14）； 10-13近场远场（12）；

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第4章 电磁波的辐射第4,5,6节第10周一（4月21日）讨论题目1： ⇒9-2辐射的多级展开

1. 在算多级展开的一阶项那部分时，为什么说微商相当于平移？
2. 磁偶极子可以近似看作一个电流圈,那么能否给出磁四级子或更高级的图像？
3. 如果系统可以看成电中性的，那么是不是可以将电极矩视为比电荷更基本的量？
4. 多级展开对应的数学基础是泰勒展开，一个数学展开与一个物理图像（球对称+偶极+四极……）对应上了，那么如果换成其他数学展开（如傅里叶展开）， 还会不会有物理的图像？
5. 我们慕课只验证了电偶极子和电四极子的电势和多级展开中的相应的一项对应，那么更高阶的怎么验证，例如电十六极矩，我根本就想象不出来它长什么样, 有没有电高阶矩等等的具体表达式？
6. 为什么我们对于四极子的模型是正方形？一般的满足要求的四边形可以吗？类似的可以推广到多极子的图像应该是什么样的？电八极子等更高阶极子 的基本模型的几何构型是什么样的？比如，电偶极子的是一个线，电四极子的是一个平行四边形。那么电八极子是否是一个平行六面体？电十六极子又如何呢？ 如果说点电荷是一个点是0维的，电偶极子是一条线是一维的，电四极子是平行四边形是二维的，那么电八极子是否是平行六面体是三维的？更高的岂不是 要变成四维甚至更高维的了？

第4章 电磁波的辐射第4,5,6节第10周周一（4月21日）讨论题目2： ⇒10-2多极矩图像

1. 讲辐射电磁场多级展开时候，对磁感应强度第k阶展开通项表达式中，偏t近似成ω原始振动的频率，怎么理解振动的源？
2. 讨论引力场的辐射的时候没有偶极辐射是因为动量守恒，动量的导数等于零，那如果体系有外力作用，动量不守恒，可能会产生偶极辐射吗？
3. 如何理解电偶极辐射能流密度最大在$\theta = \pi/2$，是哑铃；电四极是$\pi/4$，是锥；这个形状不太能直观想象。能否有什么进一步解释？还是 不太清楚为什么辐射场的一阶项就是电偶极子辐射，为什么点电荷没有贡献？
4. 计算电流圈的磁偶极辐射时，感觉最后求得矢量e_z和n的夹角是球坐标系下的θ，那辐射角分布中对θ的依赖是否都是 球坐标系意义下的θ?
5. 辐射电磁场的具体推导步骤，尤其是在推导辐射磁感应强度B_1表达式时，怎么知道每一步应该往什么方向去化？
6. 辐射场的多级展开可以写成其他形式吗？比如球谐函数。
7. 一个计算上的问题：在计算第一阶辐射场时（电四极子辐射和磁偶极子辐射项），是如何想到将x_iJ_j 改写为$x_iJ_j + x_jJ_i + x_iJ_j - x_jJ_i$然后去做计算的？这一步背后有什么更加普适的想法，还是只是一个"巧合"？

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第4章 电磁波的辐射第4,5,6节第10周三（4月23日）讨论题目： ⇒10-8卷积

1. 把场点写成卷积的形式，数学上好理解，但是对应的物理图像是什么？相当于我们把研究的问题做了什么样的转化？为什么做了这一步变换到 最后就真的能算出来空间每一点对这个点的影响？（包括之前用格林函数的时候也涉及了这个问题）

2. 这个式子里后面写的u对时间的依赖是t-R / c，可以理解为推迟效应，为什么把u(r,t)化成面上的积分时，非要写成推迟时间t*=t-R/c形式呢？ 但是如果单纯从数学的角度，可以写成u(r',t)或者u(r',t-R2/c)等，这样算会有什么结果呢，是否就不能化成全微商项？如果在最开始推导时把推迟势 换成提前势（即t*中变号），会有什么影响么（推了一遍应该没有影响）？为什么用推迟效应考虑就能化成全微商呢？刚开始写u(r,t)的时候就搞成了带 推迟的形式u(r,t-R/c），感觉不好理解，能不能写一个稍微具体一点的式子看一下这样写函数在公式上代表什么呢？

3. 推导满足波动方程的函数在某一点的值，能化为面积分的形式时，用了一些不直接的形式（如取R为r'-r），整个推导就像是知道答案以后 示了一遍，能否讲解一下，实际理论物理工作中这样的演算是如何完成的？是猜一个形式代进去狂算吗？

4. 如果把延迟势换成提前势有什么影响，这是否可以看作量子力学里面的"量子延迟"？惠更斯原理所认为的波只能向前传播，如何理解？





从讨论表面上和区域外的源分布对电磁场的贡献到关注区域内源体分布的贡献

一团荷的分布 ≡ 球对称分布 + 偶极分布 + 四极子分布 + · · ·

一团电荷在远处的电学效应

≡ 一个点电荷 + 一个电偶极子 + 一个电四极子 + · · ·
在远处的电学效应

一团电流在远处的磁学效应

≡ 一个磁偶极子 + 一个磁四极子 + · · · 在远处的磁学效应









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$g(x')$ 在$x'=0$处展开:

$$g(x') = g(0) + \sum_{n=1}^{\infty} \frac{1}{n!} \frac{\partial^n g(x')}{\partial x'^n}\Big|_{x'=0} x'^n$$

$f(\vec{r}')$ 在$\vec{r}'=0$处展开:

$$f(\vec{r}') = f(0) + \sum_{n=1}^{\infty} \sum_{i_1 i_2 \ldots i_n=1}^{3} \frac{1}{n!} \Big[\frac{\partial}{\partial x'_{i_1}} \frac{\partial}{\partial x'_{i_2}} \cdots \frac{\partial}{\partial x'_{i_n}} f(\vec{r}')\Big]_{\vec{r}'=0} x'_{i_1} x'_{i_2} \cdots x'_{i_n}$$

$$= f(0) + \sum_{n=1}^{\infty} \frac{1}{n!} \Big[(\vec{r}' \cdot \nabla'')^n \, f(\vec{r}'')\Big]_{\vec{r}''=0}$$

$$\frac{1}{|\vec{r}-\vec{r}'|} = \frac{1}{r} + \sum_{n=1}^{\infty} \sum_{i_1 i_2 \ldots i_n=1}^{3} \frac{1}{n!} \Big[\frac{\partial}{\partial x'_{i_1}} \frac{\partial}{\partial x'_{i_2}} \cdots \frac{\partial}{\partial x'_{i_n}} \frac{1}{|\vec{r}-\vec{r}'|}\Big]_{\vec{r}'=0} x'_{i_1} x'_{i_2} \cdots x'_{i_n}$$

$$= \frac{1}{r} + \sum_{n=1}^{\infty} \sum_{i_1 i_2 \ldots i_n=1}^{3} \frac{(-1)^n}{n!} \Big[\frac{\partial}{\partial x_{i_1}} \frac{\partial}{\partial x_{i_2}} \cdots \frac{\partial}{\partial x_{i_n}} \frac{1}{|\vec{r}-\vec{r}'|}\Big]_{\vec{r}'=0} x'_{i_1} x'_{i_2} \cdots x'_{i_n}$$

$$= \frac{1}{r} + \sum_{n=1}^{\infty} \sum_{i_1 i_2 \ldots i_n=1}^{3} \frac{(-1)^n}{n!} \Big[\frac{\partial}{\partial x_{i_1}} \frac{\partial}{\partial x_{i_2}} \cdots \frac{\partial}{\partial x_{i_n}} \frac{1}{r}\Big] x'_{i_1} x'_{i_2} \cdots x'_{i_n}$$

$$= \frac{1}{r} + \sum_{n=1}^{\infty} 4\pi \sum_{m=-n}^{n} \frac{1}{2n+1} \frac{r'^n}{r^{n+1}} Y^*_{nm}(\theta', \phi') Y_{nm}(\theta, \phi)$$



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$$\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int d\tau' \frac{\rho(\vec{r}')}{|\vec{r}-\vec{r}'|}$$

$$= \frac{1}{4\pi\epsilon_0}\{\frac{1}{r}\int d\tau' \rho(\vec{r}') + \sum_{n=1}^{\infty}\sum_{i_1 i_2 \ldots i_n=1}^{3} \frac{(-1)^n}{n!}[\frac{\partial}{\partial x_{i_1}}\frac{\partial}{\partial x_{i_2}}\cdots\frac{\partial}{\partial x_{i_n}}\frac{1}{r}]$$

$$\times \int d\tau' x'_{i_1} x'_{i_2} \cdots x'_{i_n} \rho(\vec{r}')\} = \sum_{n=0}^{\infty}\phi_n(\vec{r})$$

$$\phi_0(\vec{r}) = \frac{1}{4\pi\epsilon_0 r}\int d\tau' \rho(\vec{r}')$$

$$\phi_n(\vec{r})\big|_{n>0} = \frac{1}{4\pi\epsilon_0}\sum_{i_1 i_2 \ldots i_n=1}^{3} \frac{(-1)^n}{n!}[\frac{\partial}{\partial x_{i_1}}\frac{\partial}{\partial x_{i_2}}\cdots\frac{\partial}{\partial x_{i_n}}\frac{1}{r}] q_{i_1 i_2 \ldots i_n}$$

$$q_{i_1 i_2 \ldots i_n} = \int d\tau' x'_{i_1} x'_{i_2} \cdots x'_{i_n} \rho(\vec{r}')$$

计算球(椭球)形分布电荷的$q_{i_1 i_2 \ldots i_n}$,说明其高阶项对电势无贡献!

$$= \frac{1}{\epsilon_0}\sum_{m=-n}^{n} \frac{1}{2n+1}\frac{1}{r^{n+1}} Y_{nm}(\theta,\phi)\int d\tau' \rho(\vec{r}') r'^n Y^*_{nm}(\theta',\phi')$$

m有$2n+1$个取值意味着对2^n阶多极矩，实际具有$2n+1$个独立分量！

$$\vec{E} = \vec{E}_0 + \vec{E}_1 + \cdots \qquad \vec{E}_n = -\nabla\phi_n \qquad \phi_{n+1}/\phi_n \sim r'/r$$





电荷:

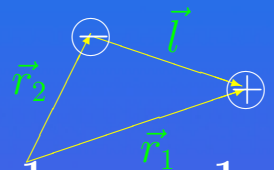
$$\phi_0(\vec{\mathbf{r}}) = \frac{1}{4\pi\epsilon_0}\frac{\mathbf{Q}}{\mathbf{r}}$$

$$\mathbf{Q} = \int \mathbf{d}\tau' \rho(\vec{\mathbf{r}}')$$

$$\vec{\mathbf{E}}_0(\vec{\mathbf{r}}) = \frac{\mathbf{Q}\vec{\mathbf{r}}}{4\pi\epsilon_0 \mathbf{r}^3}$$

体系被缩成一个电量为Q的点电荷, 体系线度的效应被略去了.

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电偶极子:

$$\phi_1(\vec{r}) = \underbrace{\frac{-1}{4\pi\epsilon_0}\sum_{i_1=1}^{3}\left[\frac{\partial}{\partial x_{i_1}}\frac{1}{r}\right]q_{i_1}}_{\text{点电荷电势平移的负号}} = \frac{1}{4\pi\epsilon_0}\sum_{i_1=1}^{3}\frac{x_{i_1}}{r^3}q_{i_1} = \frac{\vec{r}\cdot\vec{p}}{4\pi\epsilon_0 r^3} = -\frac{1}{4\pi\epsilon_0}\vec{p}\cdot\nabla\frac{1}{r}$$

$$\nabla\nabla\frac{1}{r} = -\frac{\vec{\vec{I}}}{r^3} + 3\frac{\vec{r}\vec{r}}{r^5} - \frac{4\pi}{3}\vec{\vec{I}}\,\delta(\vec{r})$$

$$\vec{p} = \sum_{i=1}^{3} q_i\vec{e}_i = \int d\tau'\vec{r}'\rho(\vec{r}') \qquad \vec{E}_1 \xrightarrow{-\frac{1}{4\pi\epsilon_0}\vec{p}\cdot\nabla\nabla\frac{1}{r}} \frac{1}{4\pi\epsilon_0}\left[-\frac{\vec{p}}{r^3} + \frac{3\vec{r}(\vec{p}\cdot\vec{r})}{r^5}\right] - \frac{\vec{p}}{3\epsilon_0}\delta(\vec{r})$$

体系被缩成一个电偶极矩为$\vec{p}$的电偶极子.一个由两个电量$\pm q$, 相距$\vec{l}$(从负指向正), 且$l \to 0, q \to \infty$, 但保持ql为有限的点电荷构成的体系. 它可看成是错位的正负电荷体系。

对此体系, 总电量$Q = q - q = 0$, 最低阶的就是偶极矩的贡献,

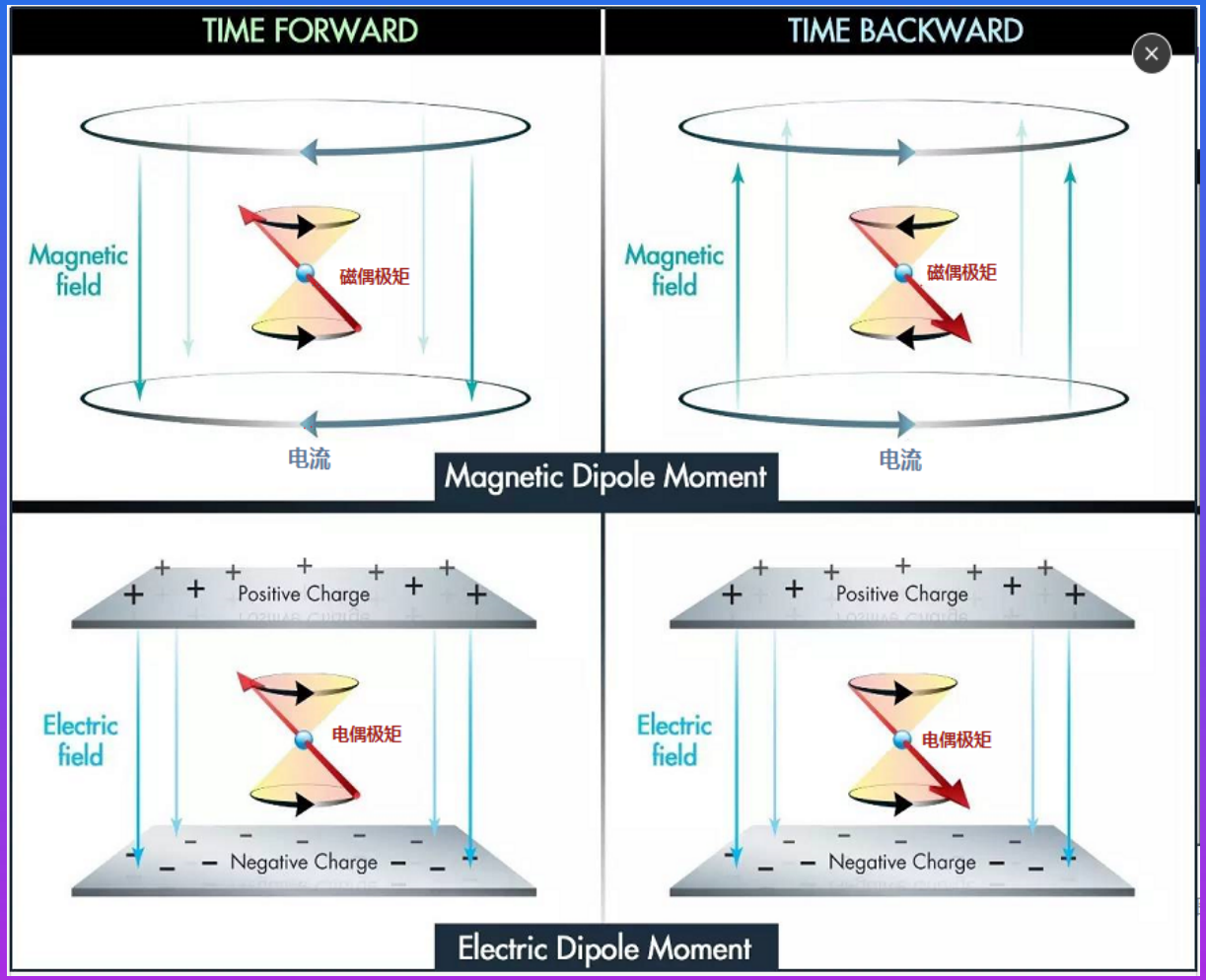
$$\vec{p} = \int d\tau'\vec{r}'\rho = q\int d\tau'\vec{r}'[\delta(\vec{r}' - \vec{r}_q) - \delta(\vec{r}' - \vec{r}_{-q})] = q(\vec{r}_q - \vec{r}_{-q}) = q\vec{l}$$

高阶项正比于 $ql(\frac{l}{r})^n \to 0(n \geq 1)$当$l \to 0$ 时趋于0.

对基本粒子：$\vec{p} = d\vec{s}$ 唯一存在的矢量 $d \neq 0$ ⇒空间反射与时间反演同时破坏电子、中子！

在空间反射变换下，p变号、自旋s不变；在时间反演变换下，p不变、自旋s变号！




TIME FORWARD
TIME BACKWARD
Magnetic field
磁偶极矩
电流
Magnetic Dipole Moment
Positive Charge
Electric field
电偶极矩
Negative Charge
Electric Dipole Moment

电四极子:

多极展开: 电势的多极展开,电多极矩

$$\phi_2(\vec{r}) = \frac{1}{2}\frac{1}{4\pi\epsilon_0}\sum_{i_1,i_2=1}^{3}[\frac{\partial}{\partial x_{i_1}}\frac{\partial}{\partial x_{i_2}}\frac{1}{r}]q_{i_1 i_2}$$

偶极子电势平移的负号；点电荷电势的二次平移

$$= \frac{1}{8\pi\epsilon_0}\sum_{i_1,i_2=1}^{3}[\frac{3x_{i_1}x_{i_2}}{r^5} - \frac{\delta_{i_1 i_2}}{r^3} - \frac{4\pi}{3}\delta_{i_1 i_2}\delta(\vec{r})]q_{i_1 i_2}$$

$$= \frac{1}{\epsilon_0}\int d\tau'\rho(\vec{r}')\left[\frac{1}{8\pi r^5}[\sum_{i_1,i_2=1}^{3}3x_{i_1}x'_{i_1}x'_{i_2}x_{i_2} - r'^2r^2] - \frac{1}{6}r'^2\delta(\vec{r})\right]$$

$$= \frac{1}{8\pi\epsilon_0 r^5}\sum_{i_1,i_2=1}^{3}x_{i_1}x_{i_2}\int d\tau'\rho[3x'_{i_1}x'_{i_2} - r'^2\delta_{i_1 i_2}] - \frac{1}{6\epsilon_0}\delta(\vec{r})\int d\tau' r'^2\rho$$

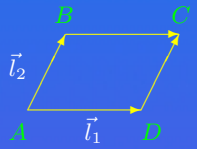
$$= \sum_{i_1,i_2=1}^{3}\frac{x_{i_1}Q_{i_1 i_2}x_{i_2}}{8\pi\epsilon_0 r^5} - \frac{1}{6\epsilon_0}\delta(\vec{r})\int d\tau' r'^2\rho = \frac{\vec{r}\cdot\overleftrightarrow{Q}\cdot\vec{r}}{8\pi\epsilon_0 r^5} - \frac{1}{6\epsilon_0}\delta(\vec{r})\int d\tau' r'^2\rho$$

$Q_{ij} = \int d\tau'[3x'_i x'_j - r'^2\delta_{ij}]\rho(\vec{r}')$ $\sum_i Q_{ii} = 0$ $\overleftrightarrow{Q} = \sum_{i,j=1}^{3}Q_{ij}\vec{e}_i\vec{e}_j = \int d\tau'[3\vec{r}'\vec{r}' - r'^2\overleftrightarrow{I}]\rho(\vec{r}')$

$$\vec{E}_2 = \frac{1}{8\pi\epsilon_0}[\frac{(\overleftrightarrow{Q}\cdot\vec{r} + \vec{r}\cdot\overleftrightarrow{Q})}{r^5} - \frac{5\vec{r}(\vec{r}\cdot\overleftrightarrow{Q}\cdot\vec{r})}{r^7}] + \frac{1}{6\epsilon_0}\int d\tau' r'^2\rho\nabla\delta(\vec{r})$$



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体系被缩成一个电四极矩为 $\overleftrightarrow{Q}$ 的 电四极子.

一个由四个放置在一个平行四边形四个顶角A B C D(边长$\vec{l}_1$ A指向D, $\vec{l}_2$ A指向B)上的分别为$q, -q, q, -q$, 且$l_1 \to 0, l_2 \to 0, q \to \infty$, 但保持$ql_1l_2$为有限的点电荷构成的体系. 它可看成是错位的正反偶极子系统。

对此体系, 总电量$Q = q - q + q - q = 0$,总电偶极矩

$\vec{p} = q\vec{r}_A - q\vec{r}_B + q\vec{r}_C - q\vec{r}_D = -q\vec{l}_2 + q\vec{l}_2 = 0$ ，最低阶是电四极矩贡献,

$$\begin{aligned}
\overleftrightarrow{Q} &= q(3\vec{r}_A\vec{r}_A - r_A^2\ \overleftrightarrow{I} - 3\vec{r}_B\vec{r}_B + r_B^2\ \overleftrightarrow{I} + 3\vec{r}_C\vec{r}_C - r_C^2\ \overleftrightarrow{I} \\
&\quad - 3\vec{r}_D\vec{r}_D + r_D^2\ \overleftrightarrow{I}) \\
&= q[3\vec{r}_A\vec{r}_A - r_A^2\ \overleftrightarrow{I} - 3(\vec{r}_A + \vec{l}_2)(\vec{r}_A + \vec{l}_2) + (\vec{r}_A + \vec{l}_2)\cdot(\vec{r}_A + \vec{l}_2)\ \overleftrightarrow{I} \\
&\quad + 3(\vec{r}_A + \vec{l}_1 + \vec{l}_2)(\vec{r}_A + \vec{l}_1 + \vec{l}_2) - (\vec{r}_A + \vec{l}_1 + \vec{l}_2)\cdot(\vec{r}_A + \vec{l}_1 + \vec{l}_2)\ \overleftrightarrow{I} \\
&\quad - 3(\vec{r}_A + \vec{l}_1)(\vec{r}_A + \vec{l}_1) + (\vec{r}_A + \vec{l}_1)\cdot(\vec{r}_A + \vec{l}_1)\ \overleftrightarrow{I}] \\
&= q(3\vec{l}_1\vec{l}_2 + 3\vec{l}_2\vec{l}_1 - 2\vec{l}_1\cdot\vec{l}_2\ \overleftrightarrow{I})
\end{aligned}$$

高阶项正比于 $ql^2(\frac{l}{r})^n \to 0 (n \geq 1)$当$l \to 0$ 时趋于0.

$$\phi(\vec{r}) = \phi(0) + \sum_{n=1}^{\infty} \sum_{i_1 i_2 \ldots i_n=1}^{3} \frac{1}{n!} \Big[\frac{\partial}{\partial x_{i_1}} \frac{\partial}{\partial x_{i_2}} \cdots \frac{\partial}{\partial x_{i_n}} \phi(\vec{r}) \Big]_{\vec{r}=0} x_{i_1} x_{i_2} \cdots x_{i_n}$$

$$= \phi(0) - \sum_{n=1}^{\infty} \frac{1}{n!} \sum_{i_1 i_2 \ldots i_n=1}^{3} x_{i_1} x_{i_2} \cdots x_{i_n} \Big[\frac{\partial}{\partial x_{i_2}} \frac{\partial}{\partial x_{i_3}} \cdots \frac{\partial}{\partial x_{i_n}} E_{i_1}(\vec{r}) \Big]_{\vec{r}=0}$$

$$\vec{E}(\vec{r}) = \vec{E}(0) + \sum_{n=1}^{\infty} \frac{1}{n!} \sum_{i_1 i_2 \ldots i_n=1}^{3} x_{i_1} x_{i_2} \cdots x_{i_n} \Big[\frac{\partial}{\partial x_{i_1}} \frac{\partial}{\partial x_{i_2}} \cdots \frac{\partial}{\partial x_{i_n}} \vec{E}(\vec{r}) \Big]_{\vec{r}=0}$$

$W = \int d\tau \rho(\vec{r}) \phi(\vec{r})$ 此式下章证明 ρ： 刚性分布

$$= \int d\tau \rho(\vec{r}) \{ \phi(0) - \sum_{n=1}^{\infty} \frac{1}{n!} \sum_{i_1 i_2 \ldots i_n=1}^{3} x_{i_1} x_{i_2} \cdots x_{i_n} \Big[\frac{\partial}{\partial x_{i_2}} \cdots \frac{\partial}{\partial x_{i_n}} E_{i_1}(\vec{r}) \Big]_{\vec{r}=0} \}$$

$$= Q\phi(0) - \sum_{n=1}^{\infty} \frac{1}{n!} \sum_{i_1 i_2 \ldots i_n=1}^{3} q_{i_1 i_2 \ldots i_n} \Big[\frac{\partial}{\partial x_{i_2}} \frac{\partial}{\partial x_{i_3}} \cdots \frac{\partial}{\partial x_{i_n}} E_{i_1}(\vec{r}) \Big]_{\vec{r}=0} = \sum_{n=0}^{\infty} W_n$$

$$W_0 = Q\phi(0); \quad W_n = -\frac{1}{n!} \sum_{i_1 i_2 \ldots i_n=1}^{3} q_{i_1 i_2 \ldots i_n} \Big[\frac{\partial}{\partial x_{i_2}} \frac{\partial}{\partial x_{i_3}} \cdots \frac{\partial}{\partial x_{i_n}} E_{i_1}(\vec{r}) \Big]_{\vec{r}=0}$$





$$W = \sum_{n=0}^{\infty} W_n$$

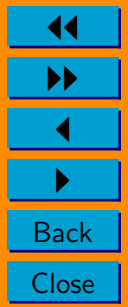
$$W_0 = Q\phi(0)$$

$$W_n = -\frac{1}{n!}\sum_{i_1 i_2 \ldots i_n=1}^{3} q_{i_1 i_2 \ldots i_n}\left[\frac{\partial}{\partial x_{i_2}}\frac{\partial}{\partial x_{i_3}}\cdots\frac{\partial}{\partial x_{i_n}}E_{i_1}(\vec{r})\right]_{\vec{r}=0}$$

$$W_1 = -\sum_{i=1}^{3} q_i E_i(0) = -\vec{p}\cdot\vec{E}(0)$$

$$W_2 = -\frac{1}{2}\sum_{i,j=1}^{3} q_{ij}\frac{\partial}{\partial x_j}E_i = -\frac{1}{6}\sum_{i,j=1}^{3}(3q_{ij} - \delta_{ij}\sum_{k=1}^{3} q_{kk})\frac{\partial}{\partial x_j}E_i$$

$$= -\frac{1}{6}\sum_{i,j=1}^{3} Q_{ij}\frac{\partial}{\partial x_j}E_i \qquad \nabla\cdot\vec{E}\big|_{\vec{r}=0} = 0$$



多极展开：带电体在外场中的能量,受力和力矩

$$\vec{F}=\int d\tau\rho(\vec{r})\vec{E}(\vec{r})$$

$$=\int d\tau\rho(\vec{r})\{\vec{E}(0)+\sum_{n=1}^{\infty}\frac{1}{n!}\sum_{i_1 i_2\ldots i_n=1}^{3}x_{i_1}x_{i_2}\cdots x_{i_n}[\frac{\partial}{\partial x_{i_1}}\cdots\frac{\partial}{\partial x_{i_n}}\vec{E}(\vec{r})]_{\vec{r}=0}\}$$

$$=Q\vec{E}(0)+\sum_{n=1}^{\infty}\frac{1}{n!}\sum_{i_1 i_2\ldots i_n=1}^{3}q_{i_1 i_2\ldots i_n}[\frac{\partial}{\partial x_{i_1}}\frac{\partial}{\partial x_{i_2}}\cdots\frac{\partial}{\partial x_{i_n}}\vec{E}(\vec{r})]_{\vec{r}=0}=\sum_{n=0}^{\infty}\vec{F}_n$$

$$\vec{F}_0=Q\vec{E}(0)\qquad \vec{F}_n=\frac{1}{n!}\sum_{i_1 i_2\ldots i_n=1}^{3}q_{i_1 i_2\ldots i_n}[\frac{\partial}{\partial x_{i_1}}\frac{\partial}{\partial x_{i_2}}\cdots\frac{\partial}{\partial x_{i_n}}\vec{E}(\vec{r})]_{\vec{r}=0}$$

$$\vec{F}_1=\sum_{i=1}^{3}q_i\frac{\partial}{\partial x_i}\vec{E}(\vec{r})\Big|_{\vec{r}=0}=\vec{p}\cdot\nabla\vec{E}(\vec{r})\Big|_{\vec{r}=0}$$

$$\vec{F}_2=\frac{1}{2}\sum_{i,j=1}^{3}q_{ij}[\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j}\vec{E}(\vec{r})]\Big|_{\vec{r}=0}=\frac{1}{6}\sum_{i,j=1}^{3}(3q_{ij}-\delta_{ij}\sum_{k=1}^{3}q_{kk})\frac{\partial^2}{\partial x_i\partial x_j}\vec{E}\Big|_{\vec{r}=0}$$

$$=\frac{1}{6}\sum_{i,j=1}^{3}Q_{ij}[\frac{\partial^2}{\partial x_i\partial x_j}\vec{E}(\vec{r})]\Big|_{\vec{r}=0}\qquad \nabla^2\vec{E}\Big|_{\vec{r}=0}=-\nabla[\nabla^2\phi]\Big|_{\vec{r}=0}=0$$



多极展开: 带电体在外场中的能量,受力和力矩

$$\vec{L} = \int d\tau \rho(\vec{r}) \vec{r} \times \vec{E}(\vec{r})$$

$$= \int d\tau \rho(\vec{r}) \vec{r} \times \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{i_1 i_2 \ldots i_n = 1}^{3} x_{i_1} \cdots x_{i_n} \left[\frac{\partial}{\partial x_{i_1}} \cdots \frac{\partial}{\partial x_{i_n}} \vec{E}(\vec{r}) \right]_{\vec{r}=0}$$

$$= \sum_{i,j,k=1}^{3} \epsilon_{ijk} \int d\tau \rho(\vec{r}) x_i \vec{e}_k \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{i_1 i_2 \ldots i_n = 1}^{3} x_{i_1} \cdots x_{i_n} \left[\frac{\partial}{\partial x_{i_1}} \cdots \frac{\partial}{\partial x_{i_n}} E_j(\vec{r}) \right]_{\vec{r}=0}$$

$$= \sum_{n=1}^{\infty} \frac{1}{n!} \sum_{i,j,k,i_1 i_2 \ldots i_n = 1}^{3} \epsilon_{ijk} q_{i i_1 \ldots i_n} \vec{e}_k \left[\frac{\partial}{\partial x_{i_1}} \cdots \frac{\partial}{\partial x_{i_n}} E_j(\vec{r}) \right]_{\vec{r}=0} = \sum_{n=0}^{\infty} \vec{L}_n$$

$$\vec{L}_n = \frac{1}{n!} \sum_{i,j,k,i_1 i_2 \ldots i_n = 1}^{3} \epsilon_{ijk} q_{i i_1 \ldots i_n} \vec{e}_k \left[\frac{\partial}{\partial x_{i_1}} \cdots \frac{\partial}{\partial x_{i_n}} E_j(\vec{r}) \right]_{\vec{r}=0}$$

$$\vec{L}_0 = \sum_{i,j,k} \epsilon_{ijk} q_i \vec{e}_k E_j(0) = \vec{p} \times \vec{E}(0) \qquad \sum_{ijk} \epsilon_{ijk} \vec{e}_k \frac{\partial}{\partial x_i} E_j = \nabla \times \vec{E} = 0$$

$$\vec{L}_1 = \sum_{i,j,k,i_1} \epsilon_{ijk} q_{i i_1} \vec{e}_k \frac{\partial}{\partial x_{i_1}} E_j \Big|_{\vec{r}=0} = \sum_{i,j,k,i_1=1}^{3} \epsilon_{ijk} \left[q_{i i_1} - \frac{\delta_{i i_1}}{3} \sum_{k=1}^{3} q_{kk} \right] \vec{e}_k \frac{\partial E_j}{\partial x_{i_1}} \Big|_{\vec{r}=0}$$



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$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int d\tau' \frac{\vec{j}(\vec{r}')}{|\vec{r}-\vec{r}'|}$$

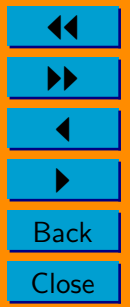
$$= \frac{\mu_0}{4\pi}\sum_{n=0}^{\infty}\sum_{i_1 i_2 \ldots i_n=1}^{3}\frac{(-1)^n}{n!}\left[\frac{\partial}{\partial x_{i_1}}\frac{\partial}{\partial x_{i_2}}\cdots\frac{\partial}{\partial x_{i_n}}\frac{1}{r}\right]\int d\tau' x'_{i_1}x'_{i_2}\cdots x'_{i_n}\vec{j}(\vec{r}')$$

$$= \sum_{n=0}^{\infty}\vec{A}_n(\vec{r})$$

$$\vec{A}_n(\vec{r}) = \frac{\mu_0}{4\pi}\sum_{i_1 i_2 \ldots i_n=1}^{3}\frac{(-1)^n}{n!}\left[\frac{\partial}{\partial x_{i_1}}\frac{\partial}{\partial x_{i_2}}\cdots\frac{\partial}{\partial x_{i_n}}\frac{1}{r}\right]\vec{J}_{i_1 i_2 \ldots i_n}$$

$$\vec{J}_{i_1 i_2 \ldots i_n} = \int d\tau' x'_{i_1}x'_{i_2}\cdots x'_{i_n}\,\vec{j}(\vec{r}')$$

$$\vec{B} = \vec{B}_0 + \vec{B}_1 + \cdots \qquad \vec{B}_n = \nabla \times \vec{A}_n$$



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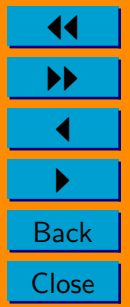
磁荷:

$$\vec{A}_0(\vec{r}) = \frac{\mu_0}{4\pi}\frac{\vec{J}}{r}$$

$$\vec{J} = \int d\tau' \vec{j}(\vec{r}') = \int d\tau' \nabla' \cdot [\vec{j}(\vec{r}')\vec{r}'] = \int_\infty d\vec{S}' \cdot [\vec{j}(\vec{r}')\vec{r}'] = 0$$

$$\vec{A}_0(\vec{r}) = 0 \qquad \vec{B}_0(\vec{r}) = 0$$

- 对稳恒的电流分布应用了电荷守恒定律，导致无磁场。
- 体系线度被略去,缩成一个点磁荷。由于无磁荷, 磁场为0。
- 对电量q位于$\vec{r}_0$的点电荷，$\vec{J} = q\dot{\vec{r}}_0 = \dot{\vec{p}}$在某处放一个静止的$-q$点电荷。



磁偶极矩:

$$\vec{A}_1(\vec{r}) = -\frac{\mu_0}{4\pi}\sum_{i_1=1}^{3}[\frac{\partial}{\partial x_{i_1}}\frac{1}{r}]\vec{J}_{i_1} = \frac{\mu_0}{4\pi}\sum_{i_1=1}^{3}\frac{x_{i_1}}{r^3}\vec{J}_{i_1} = \frac{\mu_0}{4\pi}\vec{m}\times\frac{\vec{r}}{r^3}$$

$$\vec{J}_i = \sum_{k=1}^{3}\vec{e}_k\int d\tau' x'_i j_k(\vec{r}') = \frac{1}{2}\sum_{k=1}^{3}\vec{e}_k\int d\tau'[(x'_i j_k - x'_k j_i) + (x'_i j_k + x'_k j_i)]$$

$$= \frac{1}{2}\sum_{k=1}^{3}\vec{e}_k\int d\tau'(x'_i j_k - x'_k j_i)$$

$$\int d\tau'(x'_i j_k + x'_k j_i) = \sum_{l=1}^{3}\int d\tau'\frac{\partial}{\partial x'_l}(j_l x'_i x'_k) = \int_{\infty} d\vec{S}'\cdot(\vec{j}x'_i x'_k) = 0$$

$$\sum_{i=1}^{3}x_i\vec{J}_i = \frac{1}{2}\sum_{i,k=1}^{3}\int d\tau'(x_i x'_i j_k - x'_k j_i x_i)\vec{e}_k = \frac{1}{2}\int d\tau'(\vec{r}\cdot\vec{r}'\vec{j} - \vec{r}'\vec{r}\cdot\vec{j})$$

$$= \frac{1}{2}\int d\tau'(\vec{r}'\times\vec{j})\times\vec{r} = \vec{m}\times\vec{r} \qquad \vec{m} = \frac{1}{2}\int d\tau'\vec{r}'\times\vec{j}$$

$$\vec{m} = \frac{1}{2}\sum_i \vec{r}'_i\times q_i\vec{v}'_i = \frac{1}{2}\sum_i\frac{q_i}{M_i}\vec{L}_i \overset{\text{若所有带电粒子荷质比相同}}{=\!=\!=\!=\!=\!=\!=\!=\!=\!=\!=\!=} \frac{q}{2M}\vec{L}$$

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磁偶极矩:

$$\vec{A}_1(\vec{r}) = \frac{\mu_0}{4\pi}\vec{m}\times\frac{\vec{r}}{r^3} \quad \vec{B}_1 = -\frac{\mu_0}{4\pi}\nabla\times(\vec{m}\times\nabla\frac{1}{r}) = \frac{\mu_0}{4\pi}(\vec{m}\cdot\nabla\nabla-\vec{m}\nabla^2)\frac{1}{r}$$
$$= \frac{\mu_0}{4\pi}[-\frac{\vec{m}}{r^3}+\frac{3\vec{r}(\vec{m}\cdot\vec{r})}{r^5}+\frac{8\pi}{3}\vec{m}\delta(\vec{r})]$$

体系被缩成一个磁偶极矩为$\vec{m}$的磁偶极子, 一个由 电流J均匀分布在面$\vec{S}$上,且$S\to 0, J\to\infty$, 但保持JS为有限的电流圈构成的体系. 对此体系, 最低阶的就是磁偶极矩的贡献,

$$\vec{m} \equiv \frac{1}{2}\int d\tau'\vec{r}'\times\vec{j} = -\frac{1}{2}\int d\tau'\vec{j}\times\vec{r}' = -\frac{1}{2}J\oint d\vec{l}'\times\vec{r}'$$
$$= -\frac{1}{2}J\int(d\vec{S}'\times\nabla')\times\vec{r}' = -\frac{1}{2}J\sum_{i,j=1}^{3}\int(d\vec{S}'\times\vec{e}_i)\times\vec{e}_j\partial'_i x'_j$$
$$= -\frac{1}{2}J\sum_{i=1}^{3}\int(d\vec{S}'\times\vec{e}_i)\times\vec{e}_i = -\frac{1}{2}J\sum_{i=1}^{3}\int[(d\vec{S}'\cdot\vec{e}_i)\vec{e}_i - d\vec{S}'(\vec{e}_i\cdot\vec{e}_i)]$$
$$= J\int d\vec{S}' = J\vec{S}$$

高阶项正比于$Jl'^n(n\geq 3, l'^2\sim S)\sim JS^{n/2} = JSS^{\frac{n-2}{2}}\to 0$它是由磁四极矩,磁八极矩...等等贡献的效应.



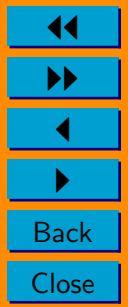
一团荷的分布 $\equiv$ 球对称分布 $+$ 偶极分布 $+$ 四极子分布 $+\cdots$

$$\mathbf{f}(\vec{\mathbf{r}}) = \mathbf{f}(\mathbf{0}) + \sum_{\mathbf{n}=1}^{\infty} \sum_{\mathbf{i}_1\mathbf{i}_2\cdots\mathbf{i}_\mathbf{n}=1}^{3} \frac{\mathbf{1}}{\mathbf{n!}} \Big[\frac{\partial}{\partial \mathbf{x}_{\mathbf{i}_1}} \frac{\partial}{\partial \mathbf{x}_{\mathbf{i}_2}} \cdots \frac{\partial}{\partial \mathbf{x}_{\mathbf{i}_\mathbf{n}}} \mathbf{f}(\vec{\mathbf{r}})\Big]_{\vec{\mathbf{r}}=\mathbf{0}} \mathbf{x}_{\mathbf{i}_1}\mathbf{x}_{\mathbf{i}_2}\cdots\mathbf{x}_{\mathbf{i}_\mathbf{n}}$$

$$= \mathbf{f}_0 + \sum_{\mathbf{n}=1}^{\infty} \sum_{\mathbf{i}_1,\cdots,\mathbf{i}_\mathbf{n}} \mathbf{f}_{\mathbf{i}_1,\cdots,\mathbf{i}_\mathbf{n}}\ \mathbf{x}_{\mathbf{i}_1}\cdots\mathbf{x}_{\mathbf{i}_\mathbf{n}}$$

$$\mathbf{f}_{\mathbf{i}_1,\cdots,\mathbf{i}_\mathbf{n}} \equiv \frac{\mathbf{1}}{\mathbf{n!}} \Big[\frac{\partial}{\partial \mathbf{x}_{\mathbf{i}_1}} \frac{\partial}{\partial \mathbf{x}_{\mathbf{i}_2}} \cdots \frac{\partial}{\partial \mathbf{x}_{\mathbf{i}_\mathbf{n}}} \mathbf{f}(\vec{\mathbf{r}})\Big]_{\vec{\mathbf{r}}=\mathbf{0}}$$

是体系的 $\mathbf{2^n}$ 阶多极距!





$$\phi(\vec{r},t)=\int d\tau'\frac{\rho(\vec{r}',t^*)}{4\pi\epsilon_0 R} \qquad \vec{A}(\vec{r},t)=\int d\tau'\frac{\mu_0\vec{j}(\vec{r}',t^*)}{4\pi R} \qquad t^*=t-\frac{R}{c}$$

$$\vec{B}=\nabla\times\vec{A} \qquad \vec{E}=-\nabla\phi-\frac{\partial\vec{A}}{\partial t}$$

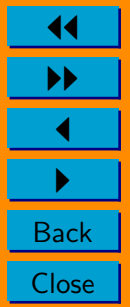
$$\oint_\infty d\vec{\sigma}\cdot\vec{S}=\oint_\infty d\vec{\sigma}\cdot(\vec{E}\times\vec{H})\doteq 0$$

$$\vec{n}\equiv\frac{\vec{R}}{R}=\frac{\vec{r}}{r}+O(\frac{1}{r}) \qquad \frac{\partial t^*}{\partial t}=1 \qquad \nabla t^*=-\nabla\frac{R}{c}=-\frac{\vec{n}}{c}$$

$$\nabla\psi(t^*)=\nabla t^*\frac{\partial\psi}{\partial t}=-\frac{\vec{n}}{c}\frac{\partial\psi}{\partial t}$$

$$\nabla\cdot\vec{f}(t^*)=\nabla t^*\cdot\frac{\partial\vec{f}}{\partial t}=-\frac{\vec{n}}{c}\cdot\frac{\partial\vec{f}}{\partial t}$$

$$\nabla\times\vec{f}(t^*)=\nabla t^*\times\frac{\partial\vec{f}}{\partial t}=-\frac{\vec{n}}{c}\times\frac{\partial\vec{f}}{\partial t}$$





$$\vec{B}_{辐} = (\nabla \times \vec{A})\big|_{\frac{1}{r}} = \int d\tau' \frac{\mu_0}{4\pi R} \nabla \times \vec{j}(\vec{r}', t^*) = \int d\tau' \frac{\mu_0}{4\pi c R} \frac{\partial \vec{j}(\vec{r}', t^*)}{\partial t} \times \vec{n}$$

$$\vec{E}_{辐} = (-\nabla\phi - \frac{\partial \vec{A}}{\partial t})\big|_{\frac{1}{r}} = \int d\tau' \Big[-\frac{1}{4\pi R \epsilon_0} \nabla \rho(\vec{r}', t^*) - \frac{\mu_0}{4\pi R} \frac{\partial \vec{j}(\vec{r}', t^*)}{\partial t} \Big]$$

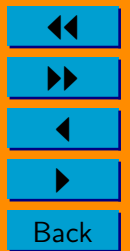
$$= \int d\tau' \Big[\frac{1}{4\pi \epsilon_0 c R} \frac{\partial \rho(\vec{r}', t^*)}{\partial t} \vec{n} - \frac{\mu_0}{4\pi R} \frac{\partial \vec{j}(\vec{r}', t^*)}{\partial t} \Big]$$

$$= \frac{\partial \phi}{\partial t} \frac{\vec{n}}{c} - \int d\tau' \frac{\mu_0}{4\pi R} \frac{\partial \vec{j}(\vec{r}', t^*)}{\partial t} = -c\vec{n} \nabla \cdot \vec{A} - \int d\tau' \frac{\mu_0}{4\pi R} \frac{\partial \vec{j}(\vec{r}', t^*)}{\partial t}$$

$$= \int d\tau' \frac{\mu_0}{4\pi R} \Big[-c\vec{n} \nabla \cdot \vec{j}(\vec{r}', t^*) - \frac{\partial \vec{j}(\vec{r}', t^*)}{\partial t} \Big]$$

$$= \int d\tau' \frac{\mu_0}{4\pi R} \Big[\vec{n} \frac{\partial \vec{j}(\vec{r}', t^*)}{\partial t} \cdot \vec{n} - \frac{\partial \vec{j}(\vec{r}', t^*)}{\partial t} \Big]$$

$$\vec{n} \times \vec{E}_{辐} = c\vec{B}_{辐} \qquad c\vec{B}_{辐} \times \vec{n} = \vec{E}_{辐}$$



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$$\vec{B}_{辐} = \int d\tau' \frac{\mu_0}{4\pi cR} \frac{\partial \vec{j}(\vec{r}', t^*)}{\partial t} \times \vec{n}$$

$$\vec{n} \times \vec{E}_{辐} = c\vec{B}_{辐} \qquad c\vec{B}_{辐} \times \vec{n} = \vec{E}_{辐}$$

$$W_e = \frac{1}{2}\epsilon_0 E^2 \qquad W_m = \frac{1}{2}\mu_0 H^2 = \frac{B^2}{2\mu_0} = \frac{E^2}{2c^2\mu_0} = \frac{1}{2}\epsilon_0 E^2 = W_e$$

$$\vec{S} = \vec{E} \times \vec{H} = c\frac{B^2}{\mu_0}\vec{n} = 2W_m c\vec{n} = (W_e + W_m)c\vec{n}$$

辐射功率:

$$I = \int_{球} d\vec{\sigma} \cdot \vec{S} = \int_{球} d\sigma \, S\vec{n} \cdot \vec{n} = \int_{球} d\sigma \, S = \int d\Omega \, Sr^2 = \int d\Omega \frac{dI}{d\Omega}$$

$$\frac{dI}{d\Omega} = Sr^2$$

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$$R = |\vec{r}-\vec{r}'| = (r^2+r'^2-2\vec{r}\cdot\vec{r}')^{\frac{1}{2}} = r(1-\vec{n}\cdot\frac{\vec{r}'}{r}+\cdots) = r-\vec{n}\cdot\vec{r}'+\cdots$$

$$\vec{A}(\vec{r},t)\big|_{辐} = \int d\tau' \frac{\mu_0}{4\pi R}\vec{j}(\vec{r}',t-\frac{R}{c}) = \int d\tau' \frac{\mu_0}{4\pi R}\vec{j}(\vec{r}',t-\frac{r}{c}+\frac{\vec{n}\cdot\vec{r}'}{c}-\cdots)$$

$$= \frac{\mu_0}{4\pi r}\int d\tau' \left[\sum_{k=0}^{\infty}\frac{1}{k!}\frac{\partial^k\vec{j}(\vec{r}',t-\frac{r}{c})}{\partial t^k}(\frac{\vec{n}\cdot\vec{r}'}{c}-\cdots)^k\right]$$

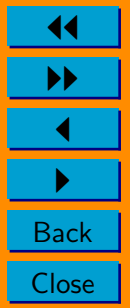
$$= \frac{\mu_0}{4\pi r}\int d\tau' \sum_{k=0}^{\infty}\frac{1}{k!}\frac{\partial^k\vec{j}(\vec{r}',t-\frac{r}{c})}{\partial t^k}(\frac{\vec{n}\cdot\vec{r}'}{c})^k$$

$$= \frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^k}\frac{\partial^k}{\partial t^k}\int d\tau'\,\vec{j}(\vec{r}',t-\frac{r}{c})x'_{i_1}\cdots x'_{i_k}$$

$$= \frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^k}\frac{\partial^k}{\partial t^k}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})$$

$$\vec{J}_{i_1\cdots i_k}(t) = \int d\tau'\vec{j}(\vec{r}',t)x'_{i_1}\cdots x'_{i_k}$$





$$\vec{J}_{i_1\cdots i_k}(t)=\int d\tau' \vec{j}(\vec{r}',t)x'_{i_1}\cdots x'_{i_k}$$

$$\vec{A}(\vec{r},t)\big|_{辐}=\frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^k}\frac{\partial^k}{\partial t^k}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})$$

$$\vec{B}(\vec{r},t)\big|_{辐}=(\nabla\times\vec{A})\big|_{\frac{1}{r}}=\frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^k}\nabla\times\frac{\partial^k}{\partial t^k}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})$$

$$=\frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^k}(-\frac{\vec{n}}{c}\times\frac{\partial}{\partial t})\frac{\partial^k}{\partial t^k}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})$$

$$=\frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^{k+1}}\frac{\partial^{k+1}}{\partial t^{k+1}}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})\times\vec{n}=\sum_{k=0}^{\infty}\vec{B}_k$$

$$\vec{B}_k=\frac{\mu_0}{4\pi r}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^{k+1}}\frac{\partial^{k+1}}{\partial t^{k+1}}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})\times\vec{n}$$

$\vec{B}_{k+1}$比$\vec{B}_k$多 $\frac{1}{c}x'\frac{\partial}{\partial t}\sim\frac{l'\omega}{c}\sim\frac{l'}{\lambda}$, 要展开好必须$l'\ll\lambda$



$$\vec{B}(\vec{r},t)\big|_{\text{辐}}=\sum_{k=0}^{\infty}\vec{B}_k \qquad \vec{B}_k=\frac{\mu_0}{4\pi r}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^{k+1}}\frac{\partial^{k+1}}{\partial t^{k+1}}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})\times\vec{n}$$

$$\vec{B}_0=\frac{\mu_0}{4\pi cr}\frac{\partial}{\partial t}\vec{J}(t-\frac{r}{c})\times\vec{n}=\frac{\mu_0}{4\pi cr}\frac{\partial}{\partial t}\int d\tau'\vec{j}(\vec{r}',t-\frac{r}{c})\times\vec{n}$$

$$=\frac{\mu_0}{4\pi cr}\sum_{i,j,k=1}^{3}\frac{\partial}{\partial t}\epsilon_{ijk}\int d\tau'\, j_i(\vec{r}',t-\frac{r}{c})n_j\vec{e}_k$$

$$=\frac{\mu_0}{4\pi cr}\sum_{i,j,k,l=1}^{3}\epsilon_{ijk}\frac{\partial}{\partial t}\int d\tau'\,\left[\partial'_l[j_l(\vec{r}',t-\frac{r}{c})x'_i]-x'_i\partial'_l j_l(\vec{r}',t-\frac{r}{c})\right]n_j\vec{e}_k$$

$$=\frac{\mu_0}{4\pi cr}\sum_{i,j,k=1}^{3}\epsilon_{ijk}\frac{\partial}{\partial t}\int d\tau'\, x'_i\frac{\partial\rho(\vec{r}',t-\frac{r}{c})}{\partial t}n_j\vec{e}_k$$

$$=\frac{\mu_0}{4\pi cr}\sum_{i,j,k=1}^{3}\epsilon_{ijk}\frac{\partial^2}{\partial t^2}q_i(t-\frac{r}{c})n_j\vec{e}_k=\frac{\mu_0}{4\pi cr}\ddot{\vec{P}}(t-\frac{r}{c})\times\vec{n}$$

$$\vec{P}(t)=\sum q_i(t)\vec{e}_i \qquad q_i(t)=\int d\tau'\, x'_i\rho(\vec{r}',t)$$

为什么没有点电荷的辐射贡献？ 实际上对点电荷第二个等号给出：$\frac{\mu_0}{4\pi cr}\ddot{q}\vec{r}(t-\frac{r}{c})\times\vec{n}$



$$\vec{B}(\vec{r},t)\big|_{\text{辐}}=\sum_{k=0}^{\infty}\vec{B}_k \qquad \vec{B}_k=\frac{\mu_0}{4\pi r}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^{k+1}}\frac{\partial^{k+1}}{\partial t^{k+1}}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})\times\vec{n}$$

$$\vec{B}_1=\frac{\mu_0}{4\pi c^2 r}\sum_{i=1}^{3}n_i\frac{\partial^2}{\partial t^2}\vec{J}_i(t-\frac{r}{c})\times\vec{n}$$

$$=\frac{\mu_0}{4\pi c^2 r}\sum_{i=1}^{3}n_i\frac{\partial^2}{\partial t^2}\int d\tau'\, x'_i\vec{j}(\vec{r}',t-\frac{r}{c})\times\vec{n}$$

$$=\frac{\mu_0}{4\pi c^2 r}\sum_{i,j,k,l=1}^{3}n_i\frac{\partial^2}{\partial t^2}\epsilon_{jkl}\int d\tau'\, x'_i j_j(\vec{r}',t-\frac{r}{c})n_k\vec{e}_l$$

$$=\frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3}n_i\frac{\partial^2}{\partial t^2}\epsilon_{jkl}\int d\tau'\,[x'_i j_j+x'_j j_i+x'_i j_j-x'_j j_i]n_k\vec{e}_l$$

$$=\frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3}n_i\frac{\partial^2}{\partial t^2}\epsilon_{jkl}\int d\tau'\Big[\sum_{m=1}^{3}[\partial'_m(j_m x'_i x'_j)-x'_i x'_j(\partial'_m j_m)]+\sum_{m,n,p=1}^{3}\epsilon_{ijp}\epsilon_{nmp}x'_n j_m\Big]n_k\vec{e}_l$$



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$$\vec{B}(\vec{r},t)\big|_{\text{辐}}=\sum_{k=0}^{\infty}\vec{B}_k \qquad \vec{B}_k=\frac{\mu_0}{4\pi r}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^{k+1}}\frac{\partial^{k+1}}{\partial t^{k+1}}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})\times\vec{n}$$

$$\vec{B}_1=\frac{\mu_0}{4\pi c^2 r}\sum_{i=1}^{3}n_i\frac{\partial^2}{\partial t^2}\vec{J}_i(t-\frac{r}{c})\times\vec{n}$$

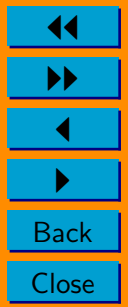
$$=\frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3}n_i\frac{\partial^2}{\partial t^2}\epsilon_{jkl}\int d\tau'\,\Big[\sum_{m=1}^{3}[\partial'_m(j_m x'_i x'_j)-x'_i x'_j(\partial'_m j_m)]$$

$$+\sum_{m,n,p=1}^{3}\epsilon_{ijp}\epsilon_{nmp}x'_n j_m\Big]n_k\vec{e}_l$$

$$=\frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3}n_i\frac{\partial^2}{\partial t^2}\int d\tau'\,\Big[\epsilon_{jkl}x'_i x'_j\frac{\partial}{\partial t}\rho(\vec{r}',t-\frac{r}{c})$$

$$+\sum_{m,n,p=1}^{3}\epsilon_{jkl}\epsilon_{ijp}\epsilon_{nmp}x'_n j_m(\vec{r}',t-\frac{r}{c})\Big]n_k\vec{e}_l$$





$$\vec{B}_1 = \frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3} n_i \frac{\partial^2}{\partial t^2}\int d\tau' \Big[\epsilon_{jkl} x'_i x'_j \frac{\partial}{\partial t}\rho(\vec{r}', t-\frac{r}{c}) + \sum_{m,n,p=1}^{3} \epsilon_{jkl}\epsilon_{ijp}\epsilon_{nmp} x'_n j_m(\vec{r}', t-\frac{r}{c})\Big] n_k \vec{e}_l$$

$$= \frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3} n_i \Big[\frac{\partial^3}{\partial t^3}\epsilon_{jkl} q_{ij}(t-\frac{r}{c}) + 2\frac{\partial^2}{\partial t^2}\sum_{p=1}^{3}\epsilon_{jkl}\epsilon_{ijp} m_p(t-\frac{r}{c})\Big] n_k \vec{e}_l$$

$$= \frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3} n_i \Big[\frac{\partial^3}{\partial t^3}\epsilon_{jkl}\frac{1}{3} Q_{ij}(t-\frac{r}{c}) + 2\frac{\partial^2}{\partial t^2}\sum_{p=1}^{3}\epsilon_{jkl}\epsilon_{ijp} m_p(t-\frac{r}{c})\Big] n_k \vec{e}_l$$

$$= \frac{\mu_0}{4\pi c^2 r}\Big[\frac{1}{6}\vec{n}\cdot \dddot{\overleftrightarrow{Q}} \times \vec{n} + (\ddot{\vec{m}}\times\vec{n})\times\vec{n}\Big]$$

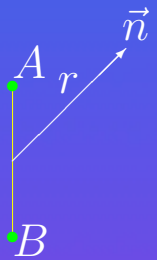
$$\overleftrightarrow{Q}(t-\frac{r}{c}) = \int d\tau'(3\vec{r}'\vec{r}' - \overleftrightarrow{I} r'^2)\rho(\vec{r}', t-\frac{r}{c}) \qquad \vec{m}(t-\frac{r}{c}) = \frac{1}{2}\int d\tau' \vec{r}'\times\vec{j}(\vec{r}', t-\frac{r}{c})$$

$\vec{B}_0 = \frac{\mu_0}{4\pi cr}\ddot{\vec{P}}(t-\frac{r}{c})\times\vec{n}$ $\qquad \vec{P}(t) = \sum q_i(t)\vec{e}_i$ $\qquad q_i(t) = \int d\tau'\, x'_i \rho(\vec{r}', t)$

若讨论引力场，$\vec{P}(t) = M\vec{R}(t)$， $\vec{R}(t)$是质心位置。$\ddot{\vec{P}}(t)$是质心动量的时间变化率由动量守恒为零！引力场无偶极辐射！

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一短天线由A,B两个小球组成,($\vec{l}$很短,由B指向A),A,B上各带相反电荷, 充电后在A B间的导线上产生高频电流



$$J = \frac{dQ_A}{dt} = J_0 \cos \omega t \qquad Q_A = \frac{J_0}{\omega} \sin \omega t$$

$$l \to 0, \ \frac{J_0}{\omega} \to \infty, \ l\frac{J_0}{\omega} \text{固定} \qquad \vec{P} = Q_A \vec{l} = \frac{J_0}{\omega} \vec{l} \sin \omega t$$

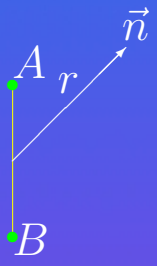
$$\vec{B}(\vec{r}, t) = \frac{\mu_0}{4\pi cr} \ddot{\vec{P}}(t - \frac{r}{c}) \times \vec{n} = -\frac{\mu_0 J_0 \omega}{4\pi cr} \vec{l} \sin \omega (t - \frac{r}{c}) \times \vec{n}$$

$$= \frac{\mu_0 \vec{l}}{4\pi cr} \dot{J}(t - \frac{r}{c}) \times \vec{n}$$

$$\vec{S}(\vec{r}, t) = \frac{c}{\mu_0} B^2 \vec{n} = \frac{\mu_0 l^2}{(4\pi)^2 c r^2} \dot{J}^2 (t - \frac{r}{c}) \sin^2 \theta \vec{n}$$

$$\frac{dI}{d\Omega} = \frac{\mu_0 l^2}{(4\pi)^2 c} \dot{J}^2 (t - \frac{r}{c}) \sin^2 \theta$$

一短天线由A,B两个小球组成,($\vec{l}$很短,由B指向A),A,B上各带相反电荷, 充电后在AB间的导线上产生高频电流



$$\frac{dI}{d\Omega}=\frac{\mu_0 l^2}{(4\pi)^2 c}\dot{J}^2(t-\frac{r}{c})\sin^2\theta$$

$$I(t)=\int d\Omega\frac{dI}{d\Omega}=\int_0^{2\pi}d\phi\int_0^{\pi}d\theta\sin\theta\frac{dI}{d\Omega}$$

$$=2\pi\int_0^{\pi}d\theta\ \frac{\mu_0 l^2}{(4\pi)^2 c}\dot{J}^2(t-\frac{r}{c})\sin^3\theta=\frac{\mu_0 l^2}{6\pi c}\dot{J}^2(t-\frac{r}{c})=\frac{\mu_0 l^2\omega^2 J_0^2}{6\pi c}\sin^2\omega(t-\frac{r}{c})$$

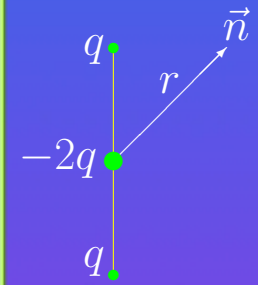
$$\bar{I}=\lim_{T\to\infty}\frac{1}{T}\int_0^T dt\ I(t)=\lim_{T\to\infty}\frac{1}{T}\int_0^T dt\ \frac{\mu_0 l^2\omega^2 J_0^2}{6\pi c}\frac{1}{2}[1-\cos 2\omega(t-\frac{r}{c})]$$

$$=\frac{\mu_0 l^2\omega^2 J_0^2}{12\pi c}\lim_{T\to\infty}\frac{1}{T}[t-\frac{1}{2\omega}\sin 2\omega(t-\frac{r}{c})]\Big|_0^T=\frac{\mu_0 l^2\omega^2 J_0^2}{12\pi c}$$

$$=\frac{\mu_0 l^2\omega^2}{6\pi cR}(\frac{1}{2}J_0^2R)=\frac{R_{辐}}{R}\ (\frac{1}{2}J_0^2R)$$

$R_{辐}=\frac{\mu_0 l^2\omega^2}{6\pi c}$　　$\frac{1}{2}J_0^2R$：　平均热功率；1/2来自时间平均　　R：　天线电阻

电量为$q, q, -2q$的三个小球分别位于长为$2l$的短天线的两端和中间,$\vec{l}$为 从天线中点指向一端的矢量,$q = q_0 \cos\omega t$, $q_0 \to \infty, l \to 0$但$q_0 l^2$固定.



$$\mathbf{J} = \dot{q} = -q_0\omega \sin\omega t = J_0 \sin\omega t$$

$$\vec{P} = \int d\tau' \vec{r}' \rho = q\vec{l} + q(-\vec{l}) = 0$$

$$\vec{m} = \frac{1}{2}\int d\tau' \vec{r}' \times \vec{j} = \frac{1}{2}\int \vec{r}' \times J d\vec{l}' = 0$$

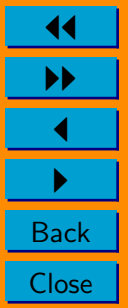
$$\overleftrightarrow{Q} = \int d\tau' 3\vec{r}'\vec{r}'\rho = 3\vec{l}\vec{l}q + 3(-\vec{l})(-\vec{l})q = 6q\vec{l}\vec{l}$$

$$\vec{B}(\vec{r}, t) = \frac{\mu_0}{6c^2 r 4\pi}\vec{n}\cdot \dddot{\overleftrightarrow{Q}} \times\vec{n} = \frac{\mu_0}{6c^2 r 4\pi} 6 \dddot{q}\, \vec{n}\cdot\vec{l}\vec{l}\times\vec{n}$$

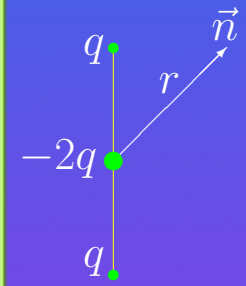
$$= \frac{l^2\mu_0 q_0 \omega^3}{4\pi c^2 r}\sin\omega(t - \frac{r}{c})\cos\theta\sin\theta\vec{e}_\phi$$

$$\vec{S}(\vec{r}, t) = \frac{c}{\mu_0}B^2\vec{n} = \frac{\mu_0 l^4 q_0^2 \omega^6}{c^3 r^2 (4\pi)^2}\sin^2\omega(t - \frac{r}{c})\sin^2\theta\cos^2\theta\vec{n}$$





电量为$q, q, -2q$的三个小球分别位于长为$2l$的短天线的两端和中间,$\vec{l}$为 从天线中点指向一端的矢量,$q = q_0 \cos\omega t$, $q_0 \to \infty, l \to 0$但$q_0 l^2$固定.



$$\vec{S}(\vec{r},t) = \frac{c}{\mu_0}B^2\vec{n} = \frac{\mu_0 l^4 q_0^2 \omega^6}{c^3 r^2 (4\pi)^2}\sin^2\omega(t-\frac{r}{c})\sin^2\theta\cos^2\theta\vec{n}$$

$$\frac{dI}{d\Omega} = \frac{\mu_0 l^4 q_0^2 \omega^6}{c^3(4\pi)^2}\sin^2\theta\cos^2\theta\sin^2\omega(t-\frac{r}{c})$$

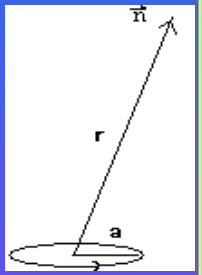
$$I = \frac{\mu_0 l^4 q_0^2 \omega^6}{c^3(4\pi)^2}\sin^2\omega(t-\frac{r}{c})2\pi\int_0^\pi d\theta\ \sin^3\theta\cos^2\theta$$

$$= \frac{\mu_0 l^4 q_0^2 \omega^6}{30c^3\pi}\sin^2\omega(t-\frac{r}{c})$$

$$\bar{I} = \frac{\mu_0 l^4 q_0^2 \omega^6}{60c^3\pi} = \frac{\mu_0 l^4 J_0^2 \omega^4}{60c^3\pi} = \frac{R_{辐}}{R}\left(\frac{1}{2}J_0^2 R\right)$$

$$R_{辐} = \frac{\mu_0 l^4 \omega^4}{30\pi c^3}$$

电流圈半径为a, $I = I_0 \cos\omega t$, $\vec{m} = I\pi a^2 \vec{e}_z$



$$\vec{j}(\vec{r},t) = j_\theta(r,z,t)\vec{e}_\theta \qquad j_r = j_z = 0$$

$$\dot{\rho}(\vec{r},t) = -\nabla\cdot\vec{j} = -\frac{1}{r}\frac{\partial j_\theta(r,z,t)}{\partial\theta} = 0 \qquad \dot{\vec{Q}} = 0$$

$$\vec{B}(\vec{r},t) = \frac{\mu_0}{4\pi c^2 r}(\ddot{\vec{m}}\times\vec{n})\times\vec{n} = -\frac{\omega^2\mu_0\pi a^2}{4\pi c^2 r} I_0 \cos\omega t^*(\vec{e}_z\times\vec{n})\times\vec{n}$$

$$\vec{S}(\vec{r},t) = \frac{c}{\mu_0}B^2\vec{n} = \frac{\omega^4\mu_0 a^4 I_0^2}{16c^3r^2}\cos^2\omega t^*[(\vec{e}_z\times\vec{n})\times\vec{n}]\cdot[(\vec{e}_z\times\vec{n})\times\vec{n}]\vec{n}$$

$$= \frac{\omega^4\mu_0 a^4 I_0^2}{16c^3r^2}\cos^2\omega t^*(\vec{e}_z\times\vec{n})\cdot(\vec{e}_z\times\vec{n})\vec{n} = \frac{\omega^4\mu_0 a^4 I_0^2}{16c^3r^2}\cos^2\omega t^*\sin^2\theta\vec{n}$$

$$\frac{dI}{d\Omega} = \frac{\omega^4\mu_0 a^4 I_0^2}{16c^3}\cos^2\omega t^*\sin^2\theta \qquad I(t) = \frac{\pi^3\omega^4\mu_0 a^4 I_0^2}{6c^3}\cos^2\omega t^*$$

$$\bar{I} = \frac{\pi^3\omega^4\mu_0 a^4 I_0^2}{12c^3} = \frac{R_{辐}}{R}\left(\frac{1}{2}J_0^2 R\right) \qquad R_{辐} = \frac{\mu_0 a^4\omega^4\pi^3}{6c^3}$$

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两个相距一定距离(不要求很远)的天线1和2，上面分别具有以同样角频率 ω 振荡的电荷电流分布，产生相应的电磁场， 叫 发射电磁波；同时它们又分别感受到对方天线所产生的电磁场，叫 接收电磁波。 发射电磁波的能量用其自身热功率度量：

$$J_i^2 R_i = J_i V_i = \int dV\ \vec{j}_i \cdot \vec{E}_i = \int dV\ \vec{v}_i \cdot (\underbrace{\rho_i \vec{E}_i + \rho_i \vec{v}_i \times \vec{B}_i}_{\text{洛伦兹力密度 } \vec{f}_i}) \qquad i = 1, 2$$

- 1 接收到的 2 发出电磁波的功率为：$P_{1a} = \int dV\ \vec{j}_1 \cdot \vec{E}_2 = \int dV\ \vec{v}_1 \cdot (\rho_1 \vec{E}_2 + \rho_1 \vec{v}_1 \times \vec{B}_2)$
- 2 接收到的 1 发出电磁波的功率为：$P_{2a} = \int dV\ \vec{j}_2 \cdot \vec{E}_1 = \int dV\ \vec{v}_2 \cdot (\rho_2 \vec{E}_1 + \rho_2 \vec{v}_2 \times \vec{B}_1)$

$$\int dV\ \vec{j}_1 \cdot \vec{E}_2 = \int dV\ \vec{j}_2 \cdot \vec{E}_1 \Rightarrow$$ 双方频率相同并感受到对方的场时，各自接收到 相同的 能量！

$$\int dV\ (\vec{j}_1 \cdot \vec{E}_2 - \vec{j}_2 \cdot \vec{E}_1) \overset{\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}}{=\!=\!=\!=\!=\!=\!=\!=} \int dV\ [(\nabla \times \vec{H}_1 - \frac{\partial \vec{D}_1}{\partial t}) \cdot \vec{E}_2 - (\nabla \times \vec{H}_2 - \frac{\partial \vec{D}_2}{\partial t}) \cdot \vec{E}_1]$$

$$\overset{\frac{\partial \vec{D}_i}{\partial t} = -i\omega \vec{E}_i}{=\!=\!=\!=\!=\!=\!=\!=} \int dV\ [\nabla \cdot (\vec{H}_1 \times \vec{E}_2 - \vec{H}_2 \times \vec{E}_1) + (\vec{H}_1 \times \nabla) \cdot \vec{E}_2 - (\vec{H}_2 \times \nabla) \cdot \vec{E}_1]$$

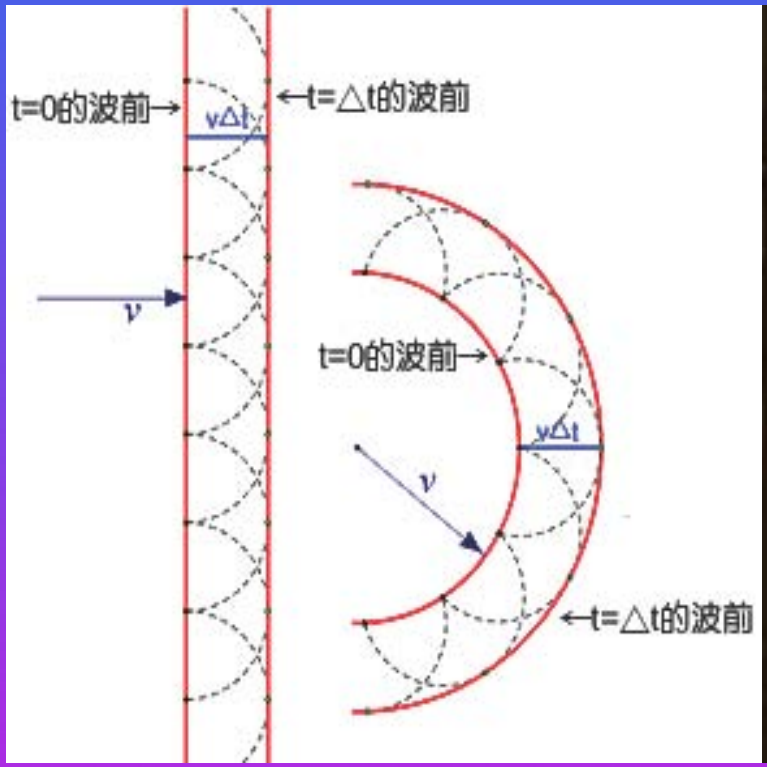
$$= \int_\infty d\vec{S} \cdot (\underbrace{\vec{H}_1 \times \vec{E}_2 - \vec{H}_2 \times \vec{E}_1}_{\vec{H}_i = \frac{1}{c\mu_0} \vec{n} \times \vec{E}_i}) + \int dV [\underbrace{(\nabla \times \vec{E}_2) \cdot \vec{H}_1 - (\nabla \times \vec{E}_1) \cdot \vec{H}_2}_{\nabla \times \vec{E}_i = i\omega\mu \vec{H}_i}]$$ 两源都在有限区用统一的 $\vec{n}$

$$= \frac{1}{c\mu_0} \int_\infty d\vec{S} \cdot [(\vec{n} \times \vec{E}_1) \times \vec{E}_2 - (\vec{n} \times \vec{E}_2) \times \vec{E}_1] + \int dV\ i\omega\mu (\vec{H}_2 \cdot \vec{H}_1 - \vec{H}_1 \cdot \vec{H}_2) \overset{\vec{n} \cdot \vec{E}_i = 0}{=\!=\!=\!=\!=} 0$$

若进一步考虑两天线之间的方向 θ, ϕ，假设各自发射都相等的总电磁波功率 P， 则：$P_{1a} = PS_2(\theta,\phi) A_1(\theta,\phi)$，$P_{2a} = PS_1(\theta,\phi) A_2(\theta,\phi)$，其中$S_i(\theta,\phi), A_i(\theta,\phi)$分别是天线i的发射和接收参数（$PS_i$是天线i的辐射角分布），天线互易原理给出$P_{1a} = P_{2a}$，得 $\frac{S_2(\theta,\phi)}{A_2(\theta,\phi)} = \frac{S_1(\theta,\phi)}{A_1(\theta,\phi)}$。

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惠更斯原理：在波传播的波阵面上的每一点都是次级波源，每一个波源发出小的球面子波,球面子波的包面形成新的波阵面.





惠更斯原理: 在波传播的波阵面上的每一点都是次级波源，每一个波源发出小的球面子波,球面子波的包面形成新的波阵面.

$$u \subset E_x, E_y, E_z, B_x, B_y, B_z$$

$$\nabla^2 u(\vec{r}, t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}u(\vec{r}, t) = 0 \qquad \vec{R} = \vec{r}' - \vec{r} \qquad \vec{r} \subset \tau$$

$$u(\vec{r}, t) = \int d\tau'\, \delta(\vec{r}' - \vec{r}) u(\vec{r}', t - \frac{R}{c}) = \int d\tau' (-\frac{1}{4\pi})(\nabla'^2 \frac{1}{R}) u(\vec{r}', t - \frac{R}{c})$$

$$= \int d\tau' (-\frac{1}{4\pi})\{\nabla' \cdot [(\nabla' \frac{1}{R}) u(\vec{r}', t - \frac{R}{c})] - (\nabla' \frac{1}{R}) \cdot \nabla' u(\vec{r}', t - \frac{R}{c})\}$$

$$= \int d\tau' (\frac{1}{4\pi})\{\nabla' \cdot [(\nabla' \frac{1}{R}) u(\vec{r}', t - \frac{R}{c}) - \frac{1}{R}\nabla' u(\vec{r}', t - \frac{R}{c})] + \frac{1}{R}\nabla'^2 u(\vec{r}', t - \frac{R}{c})\}$$

$$\nabla' t^* = -\frac{1}{c}\nabla' R = -\frac{\vec{R}}{cR}$$

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$$\frac{1}{R}\nabla'^2 u(\vec{r}', t-\frac{R}{c}) = \frac{1}{R}\nabla'\cdot[\nabla' u(\vec{r}', t^*)\big|_{t^*} - \frac{\partial u(\vec{r}', t^*)}{\partial t^*}\frac{\vec{R}}{cR}]$$

$$= \frac{1}{R}\nabla'\cdot[\nabla' u(\vec{r}', t^*)\big|_{t^*}] - \frac{\vec{R}}{cR^2}\cdot\nabla'\frac{\partial u(\vec{r}', t^*)}{\partial t^*} - \frac{1}{cR}\frac{\partial u(\vec{r}', t^*)}{\partial t^*}\nabla'\cdot\frac{\vec{R}}{R}$$

$$= \frac{1}{R}\nabla'^2 u(\vec{r}', t^*)\big|_{t^*} - \frac{\vec{R}}{cR^2}\cdot\frac{\partial}{\partial t^*}\nabla' u(\vec{r}', t^*)\big|_{t^*} - \frac{\vec{R}}{cR^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}', t^*)\big|_{t^*}$$

$$+\frac{1}{c^2 R}\frac{\partial^2}{\partial t^{*2}}u(\vec{r}', t) - \frac{\partial u(\vec{r}', t^*)}{\partial t^*}\frac{1}{cR}[\frac{3}{R} - \frac{1}{R}]$$

$$= \frac{1}{R}[\nabla'^2 u(\vec{r}', t^*)\big|_{t^*} + \frac{1}{c^2}\frac{\partial^2}{\partial t^{*2}}u(\vec{r}', t^*)] - \frac{2\vec{R}}{cR^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}', t^*)\big|_{t^*}$$

$$-\frac{2}{cR^2}\frac{\partial u(\vec{r}', t)}{\partial t^*}$$

$$= \frac{2}{Rc^2}\frac{\partial^2}{\partial t^{*2}}u(\vec{r}', t^*) - \frac{2\vec{R}}{cR^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}', t^*)\big|_{t^*} - \frac{2}{cR^2}\frac{\partial u(\vec{r}', t)}{\partial t^*}$$

$$= \frac{2}{c}\{\frac{\vec{R}}{cR}\cdot\frac{\vec{R}}{R^2}\frac{\partial^2}{\partial t^{*2}}u(\vec{r}', t^*) - \frac{\vec{R}}{R^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}', t^*)\big|_{t^*} - (\nabla'\cdot\frac{\vec{R}}{R^2})\frac{\partial u(\vec{r}', t^*)}{\partial t^*}\}$$

$$\frac{1}{R}\nabla'^2 u(\vec{r}', t-\frac{R}{c})$$

$$= \frac{2}{c}\{\frac{\vec{R}}{cR}\cdot\frac{\vec{R}}{R^2}\frac{\partial^2}{\partial t^{*2}}u(\vec{r}', t^*) - \frac{\vec{R}}{R^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}', t^*)\big|_{t^*} - (\nabla'\cdot\frac{\vec{R}}{R^2})\frac{\partial u(\vec{r}', t^*)}{\partial t^*}\}$$

$$= -\frac{2}{c}\{\frac{\vec{R}}{R^2}\cdot[-\frac{\vec{R}}{cR}\frac{\partial}{\partial t^*} + \nabla'\big|_{t^*}]\frac{\partial}{\partial t^*}u(\vec{r}', t^*) + (\nabla'\cdot\frac{\vec{R}}{R^2})\frac{\partial u(\vec{r}', t^*)}{\partial t^*}\}$$

$$= -\frac{2}{c}\{\frac{\vec{R}}{R^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}', t^*) + (\nabla'\cdot\frac{\vec{R}}{R^2})\frac{\partial u(\vec{r}', t^*)}{\partial t^*}\} = -\frac{2}{c}\nabla'\cdot\{\frac{\vec{R}}{R^2}\frac{\partial}{\partial t^*}u(\vec{r}', t^*)\}$$

$$u(\vec{r}, t) = (-\frac{1}{4\pi})\int_\tau d\tau'\ \nabla'\cdot\{(\nabla'\frac{1}{R})u(\vec{r}', t-\frac{R}{c}) - \frac{1}{R}\nabla' u(\vec{r}', t-\frac{R}{c})$$

$$-\frac{2}{c}\frac{\vec{R}}{R^2}\frac{\partial}{\partial t^*}u(\vec{r}', t^*)\}$$

$$= (-\frac{1}{4\pi})\int_\tau d\tau'\nabla'\cdot\{-\frac{\vec{R}}{R^3}u(\vec{r}', t-\frac{R}{c}) - \frac{1}{R}\nabla' u(\vec{r}', t^*)\big|_{t^*} - \frac{\vec{R}}{cR^2}\frac{\partial}{\partial t^*}u(\vec{r}', t^*)\}$$

$$= \frac{1}{4\pi}\int_\tau d\vec{S}'\ \cdot\{\frac{1}{R}\nabla' u(\vec{r}', t^*) + \frac{\vec{R}}{R^3}u(\vec{r}', t^*) + \frac{\vec{R}}{cR^2}\frac{\partial}{\partial t^*}u(\vec{r}', t^*)\}\big|_{t^*=t-\frac{R}{c}}$$

$$= \frac{1}{4\pi}\int_\tau dS'\frac{\tilde{\sigma}(\vec{r}', \vec{R}, t^*)}{R} \qquad \tilde{\sigma}(\vec{r}, \vec{R}, t) \equiv \vec{n}'\cdot\nabla u(\vec{r}, t) + \frac{\vec{n}'\vec{R}}{R^2}u(\vec{r}, t) + \frac{\vec{n}'\vec{R}}{cR}\frac{\partial}{\partial t}u(\vec{r}, t)$$



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考虑平面波$\mathbf{u}(\vec{\mathbf{r}},\mathbf{t})$ 射向开有小孔的无穷大平面S上.作如下基尔霍夫假设:

- 除孔内外,$u,\nabla u$在S面上处处为0.
- 孔内的$u,\nabla u$的值等于没有任何屏或障碍物时,入射波的值.

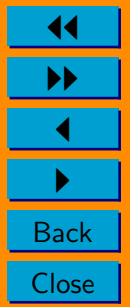
$$\mathbf{u}(\vec{\mathbf{r}},\mathbf{t})=\begin{cases}0 & \text{孔外}\\ \mathbf{A}\mathbf{e}^{\mathbf{i}(\omega\mathbf{t}-\frac{\omega\mathbf{z}}{\mathbf{c}})} & \text{孔内}\end{cases}$$

$$\mathbf{u}(\vec{\mathbf{r}},\mathbf{t})=\frac{1}{4\pi}\int_{\tau}\mathbf{d}\vec{\mathbf{S}}'\cdot\{\frac{1}{\mathbf{R}}\nabla'\mathbf{u}(\vec{\mathbf{r}}',\mathbf{t}^*)+\frac{\vec{\mathbf{R}}}{\mathbf{R}^3}\mathbf{u}(\vec{\mathbf{r}}',\mathbf{t}^*)+\frac{\vec{\mathbf{R}}}{\mathbf{cR}^2}\frac{\partial}{\partial\mathbf{t}^*}\mathbf{u}(\vec{\mathbf{r}}',\mathbf{t}^*)\}\Big|_{\mathbf{t}^*=\mathbf{t}-\frac{\mathbf{R}}{\mathbf{c}}}$$

$$\nabla'\mathbf{u}(\vec{\mathbf{r}}',\mathbf{t}^*)=-\frac{\mathbf{i}\omega}{\mathbf{c}}\vec{\mathbf{e}}_{\mathbf{z}}\mathbf{u}(\vec{\mathbf{r}}',\mathbf{t}^*)\qquad\frac{\partial}{\partial\mathbf{t}^*}\mathbf{u}(\vec{\mathbf{r}}',\mathbf{t}^*)=\mathbf{i}\omega\mathbf{u}(\vec{\mathbf{r}}',\mathbf{t}^*)$$

$$\mathbf{u}(\vec{\mathbf{r}},\mathbf{t})=\frac{1}{4\pi}\int_{\text{孔}}\mathbf{d}\vec{\mathbf{S}}'\cdot\{\frac{1}{\mathbf{R}}(-\frac{\mathbf{i}\omega}{\mathbf{c}}\vec{\mathbf{e}}_{\mathbf{z}})+\frac{\vec{\mathbf{R}}}{\mathbf{R}^3}+\frac{\vec{\mathbf{R}}}{\mathbf{cR}^2}\mathbf{i}\omega\}\mathbf{u}(\vec{\mathbf{r}}',\mathbf{t}^*)\Big|_{\mathbf{t}^*=\mathbf{t}-\frac{\mathbf{R}}{\mathbf{c}}}$$

$$=\frac{\mathbf{A}}{4\pi}\int_{\text{孔}}\mathbf{d}\vec{\mathbf{S}}'\cdot\{\frac{1}{\mathbf{R}}(-\frac{\mathbf{i}\omega}{\mathbf{c}}\vec{\mathbf{e}}_{\mathbf{z}})+\frac{\vec{\mathbf{R}}}{\mathbf{R}^3}+\frac{\vec{\mathbf{R}}}{\mathbf{cR}^2}\mathbf{i}\omega\}\mathbf{e}^{\mathbf{i}\omega\mathbf{t}^*}\Big|_{\mathbf{t}^*=\mathbf{t}-\frac{\mathbf{R}}{\mathbf{c}}}$$





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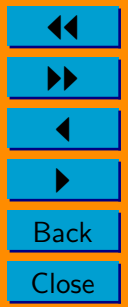
$$\mathbf{u(\vec{r},t)} = \begin{cases} 0 & \text{孔外} \\ \mathbf{Ae^{i(\omega t - \frac{\omega z}{c})}} & \text{孔内} \end{cases}$$

$$\mathbf{u(\vec{r},t)} = \frac{\mathbf{A}}{4\pi}\int_{\text{孔}} \mathbf{d\vec{S}'}\cdot\{\frac{1}{R}(-\frac{i\omega}{c}\vec{e}_z)+\frac{\vec{R}}{R^3}+\frac{\vec{R}}{cR^2}i\omega\}e^{i\omega t^*}\Big|_{t^*=t-\frac{R}{c}}$$

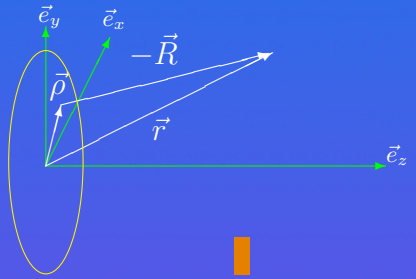
$$\overset{\mathbf{d\vec{S}'=-dS'\vec{e}_z}}{=\!=\!=\!=\!=} -\frac{\mathbf{A}}{4\pi}e^{i\omega t}\int_{\text{孔}}\mathbf{dS'}\{\frac{-i\omega}{cR}+\frac{-\cos\theta}{R^2}-\frac{i\omega}{cR}\cos\theta\}e^{-i\frac{\omega R}{c}}$$

$$= \frac{\mathbf{A}}{4\pi}e^{i\omega t}\int_{\text{孔}}\mathbf{dS'}\{\frac{i\omega}{cR}(1+\cos\theta)+\frac{\cos\theta}{R^2}\}e^{-i\frac{\omega R}{c}}$$

- 当场点离屏较远时,主要第一项贡献,叫夫琅和费衍射
- 当场点离屏较近时,主要第二项贡献,叫菲涅尔衍射



对一个半径为a的圆形小孔：



$$\vec{\rho} = \rho\cos\phi\vec{e}_{\parallel} + \rho\sin\phi\vec{e}_{\perp}$$

$$-\vec{R} = \vec{r} - \vec{\rho} \qquad \vec{r}\cdot\vec{e}_z = r\cos\theta_3$$

$$\vec{r}\cdot\vec{e}_{\parallel} = r\sin\theta_3 \qquad \vec{r}\cdot\vec{e}_{\perp} = 0$$

$$R = (r^2 + \rho^2 - 2\vec{r}\cdot\vec{\rho})^{\frac{1}{2}} = [r^2 + \rho^2 - 2r\rho\cos\phi\sin\theta_3]^{\frac{1}{2}}$$

$$\cos\theta = \frac{-\vec{R}\cdot\vec{e}_z}{R} = \frac{r\cos\theta_3}{R} = \frac{r\cos\theta_3}{[r^2 + \rho^2 - 2r\rho\cos\phi\sin\theta_3]^{\frac{1}{2}}}$$

$$R = r[1 + \frac{\rho^2}{r^2} - \frac{2\rho}{r}\cos\phi\sin\theta_3]^{\frac{1}{2}}$$

$$\overset{\rho\ll r}{=\!=\!=} r - \rho\cos\phi\sin\theta_3 + \cdots$$

$$\cos\theta = \cos\theta_3[1 + \frac{\rho}{r}\cos\phi\sin\theta_3 + \cdots]$$

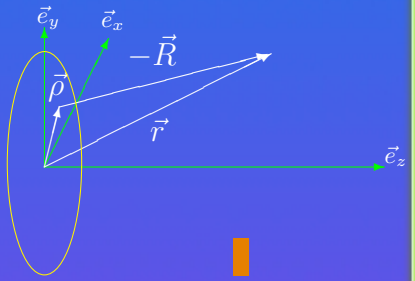
$$u(\vec{r},t) = \frac{A}{4\pi}e^{i\omega t}\int_{\text{孔}} dS'\{\frac{i\omega}{cR}(1+\cos\theta) + \frac{\cos\theta}{R^2}\}e^{-i\frac{\omega R}{c}}$$

$$= \frac{A}{4\pi}e^{i\omega t}\int_0^a \rho d\rho\int_0^{2\pi} d\phi\frac{i\omega}{cr}(1+\cos\theta_3)e^{-i\frac{\omega}{c}[r-\rho\cos\phi\sin\theta_3]}$$

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对一个半径为a的圆形小孔:

$$\mathbf{J_0(z)} = \frac{1}{2\pi}\int_0^{2\pi} \mathbf{d}\phi\ \mathbf{e}^{\mathbf{iz}\cos\phi} \qquad \frac{\mathbf{d}}{\mathbf{dz}}[\mathbf{zJ_1(z)}] = \mathbf{zJ_0(z)}$$



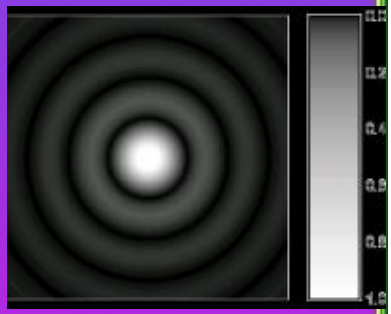
$$\mathbf{u}(\vec{\mathbf{r}},\mathbf{t}) = \frac{\mathbf{A}}{4\pi}\mathbf{e}^{\mathbf{i}\omega\mathbf{t}}\int_{\text{孔}}\mathbf{dS}'\{\frac{\mathbf{i}\omega}{\mathbf{cR}}(1+\cos\theta)+\frac{\cos\theta}{\mathbf{R^2}}\}\mathbf{e}^{-\mathbf{i}\frac{\omega\mathbf{R}}{\mathbf{c}}}$$

$$= \frac{\mathbf{A}}{4\pi}\mathbf{e}^{\mathbf{i}\omega\mathbf{t}}\frac{\mathbf{i}\omega}{\mathbf{cr}}(1+\cos\theta_3)\int_0^{\mathbf{a}}\rho\mathbf{d}\rho\int_0^{2\pi}\mathbf{d}\phi\ \mathbf{e}^{-\mathbf{i}\frac{\omega}{\mathbf{c}}[\mathbf{r}-\rho\cos\phi\sin\theta_3]}$$

$$= \frac{\mathbf{A}}{2}\mathbf{e}^{\mathbf{i}\omega(\mathbf{t}-\frac{\mathbf{r}}{\mathbf{c}})}\frac{\mathbf{i}\omega}{\mathbf{cr}}(1+\cos\theta_3)\int_0^{\mathbf{a}}\rho\mathbf{d}\rho\ \mathbf{J_0}(\frac{\rho\omega}{\mathbf{c}}\sin\theta_3)$$

$$= \mathbf{e}^{\mathbf{i}\omega(\mathbf{t}-\frac{\mathbf{r}}{\mathbf{c}})}\frac{\mathbf{icA}}{\mathbf{2\omega r}\sin\theta_3^2}(1+\cos\theta_3)\int_0^{\frac{\mathbf{a}\omega}{\mathbf{c}}\sin\theta_3}\mathbf{dz}\ \mathbf{zJ_0(z)}$$

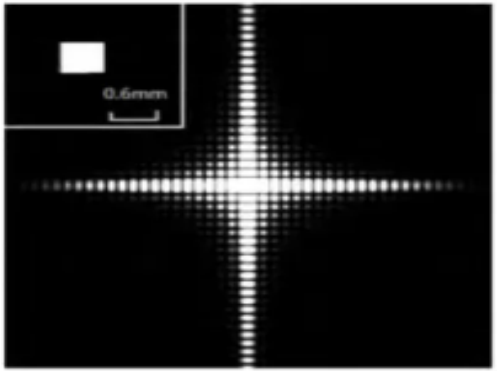
$$= \mathbf{e}^{\mathbf{i}\omega(\mathbf{t}-\frac{\mathbf{r}}{\mathbf{c}})}\frac{\mathbf{iaA}}{\mathbf{2r}}(1+\cos\theta_3)\frac{\mathbf{J_1}(\frac{\mathbf{a}\omega}{\mathbf{c}}\sin\theta_3)}{\sin\theta_3}$$



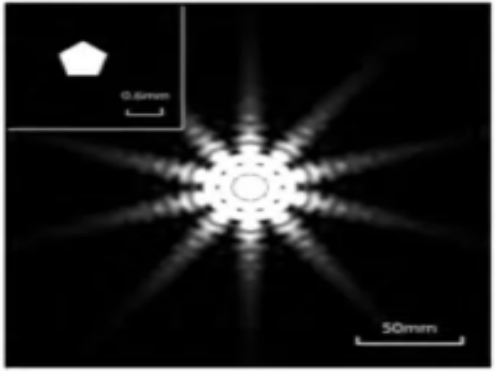
衍射波为球面波,振幅在$\frac{\omega}{\mathbf{c}}\mathbf{a}\sin\theta_3 = \mathbf{z_n}, (\mathbf{n}=1,2,3)$处为0(暗点).

$z_1 = 3.83,\ z_2 = 7.02,\ z_3 = 10.2,\ z_4 = 13.3,\ z_5 = 16.5,\ z_6 = 19.6$

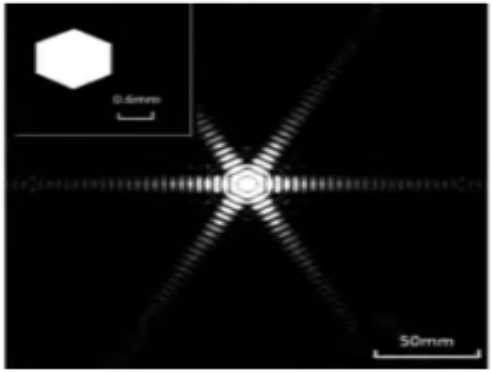
Back
Close



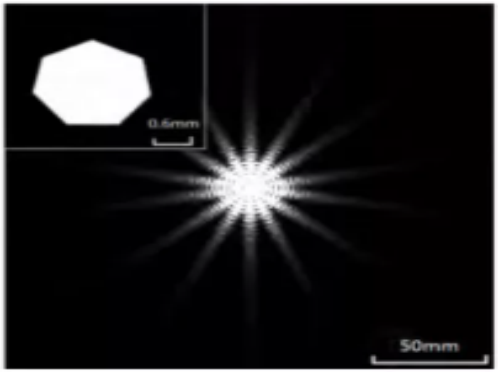
（a）正方形孔



（b）正五边形孔



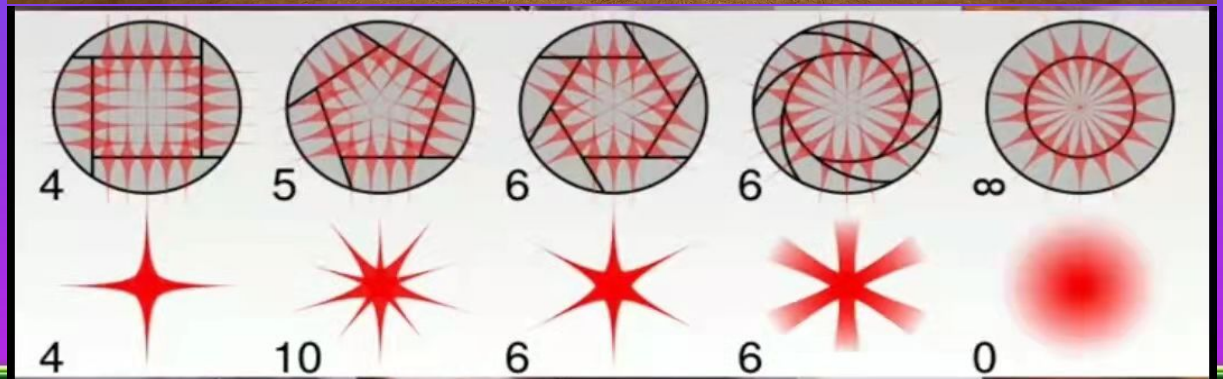
（c）正六边形孔



（d）正七边形孔

耀眼的光芒：正多边形孔衍射图样-偶数边偶数光刺、奇数边两倍个光刺”




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